

GABRIEL BAZALAR

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EDUCATION

Purdue University

Bachelor of Science in Electrical Engineering (BSEE)

Concentrations: Microelectronics & Semiconductors, and Automatic Controls

Relevant Coursework: Microprocessors Interfacing, Digital Systems Verilog and FPGA, Feedback-Controls Design,

Electromechanical Devices, Signal and Systems Analysis, Advanced C Programming, Python for Data Science

West Lafayette, IN

Expected May 2026

RESEARCH AND PROFESSIONAL EXPERIENCE

Flex Laboratory, Purdue Discovery Park

Undergraduate Researcher – Piezoelectric Sensors

West Lafayette, IN

May 2024 – Present

- Analyzed and assembled piezoelectric harvesting circuits for over 100 ceramic and polymer sensors, applying impedance matching/analysis techniques to ensure data integrity across a range of 5 kHz to 1 MHz.
- Applied a Random Forest Regressor model in Python to predict concrete strength through piezoelectric impedance analysis, enhancing the laboratory's model forecast by 15% using a Pearson correlation matrix for chemical compositions.

Calite Inmobiliaria, Gerencia y Construcción

Hardware Engineering Intern

Lima, Perú

April 2023 – August 2023

- Deployed over 45 IoT infrastructures such as networks in Ubiquity UniFi and tested microcontroller circuits for optimal performance of security cameras, intercoms, and elevator communication systems.
- Implemented a 3D-AR visualization tool in Unity Engine for Revit models, enhancing collaboration and understanding among more than 50 architects, structural engineers, and project managers.

EXTRACURRICULAR

President of Latin Autonomous Projects club (LAPs)

May 2024 – Present

- Founded an engineering club to encourage Latino students' participation in autonomous systems, fostering diversity and innovation in Machine Learning, System Controls, and Embedded Hardware at Purdue University.
- Led the collaboration LAPSxEVC to build a self-driving kart for the EV Grand Prix competition, designing custom PID control systems and interface modules with STM32 NUC, Sevcon controllers, and a Jetson TX2.

Purdue Electric Formula Racing club (PER)

August 2022 – August 2023

- Worked on the Charging Shutdown development for the BMS Team, utilizing Altium Designer for PCB layouts and participating in purchasing processes to optimize worth of component cost and quality.
- Attended workshops on digital communications protocols, specifically SPI, I2C, and CAN Bus for sending and receiving data for emergency brake or battery information.

INVOLVEMENT AND LEADERSHIP

ECE Student Ambassadors

January 2024 – Present

- Organized and represented events, contests, and tours for the College of Electrical Engineering, managing a semester budget exceeding \$8,000 to fund student recruitment, innovative projects, and student-led initiatives.

Undergraduate Teacher Assistant

January 2024 – August 2024

- Simulated circuit behaviors using LTSpice and KiCad to model and analyze RLC signal stability, CMOS behavior, and Operational Amplifiers, improving learning experience and promoting practical engagement among students.

RELEVANT PROJECTS

STM32 Video Game Console

October 2024

- Emulated the video game Piano Tiles on an STM32, displaying the game on a TFT display and controlling it with a joystick.
- Implemented an EEPROM, SD card reader, audio amplifier, and 7-segment displays for full console experience.

FPGA Lunar Lander Project

April 2024

- Designed a Verilog Lunar Lander simulator, displaying altitude, velocity, thrust, and fuel on 7-segment displays.
- Developed modules for arithmetic operations (Binary Coded Decimal), finite state machine (FSM), and input decoding.

SKILLS AND INTERESTS

Programming: C, C++ (ROS2), SystemVerilog, RISC-V Assembly, Python (TensorFlow), MATLAB, JavaScript (React.js)

CAD Tools: Altium Designer, LTSpice, KiCad, AutoCAD, Simulink, 3DsMax, Unity, Fusion 360

Certifications: IBM – ML with TensorFlow, NVIDIA – AI on Jetson Nano, IPA'D – Advanced 3DsMax

Interests: 3D Modeling, Sim Racing video games, Weightlifting, Tennis, Piano, Peruvian Acoustic Guitar